

OICPC455

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Machine Learning Lab

Quality Laboratory Manual

Experiment No. 8

Study and Implementation of Association Rule Mining



Course Instructor -
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Experiment No. 8

Title of Experiment: To study and implement Association Rule Mining

Aim of Experiment: To implement and understand the working of association rule mining algorithm, used to discover interesting relationships between variables in large datasets in Machine Learning

System Requirements – Win 8 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Needed for Experiment – Jupyter Notebook/ Anaconda Navigator/ Google Colaboratory/ Spyder, Python 3.x [With libraries such as Numpy, Pandas, Matplotlib and Scikit-Learn]

Experiment Outcomes –

1. Understand the concept and fundamentals of association rule mining algorithm and its relevance to unsupervised learning
2. To understand the importance of discovering meaningful patterns and correlations from transactional data
3. To apply the algorithms to identify frequent itemsets and generate association rules
4. To calculate and interpret the key metrics used in association rule mining

Theory –

The Apriori Algorithm, used for the first phase of the Association Rules, is the most popular and classical algorithm in the frequent item sets. These algorithm properties and data are evaluated with Boolean Association Rules. In this algorithm, there are product clusters that pass frequently, and then strong relationships between these products and other products are sought.

The importance of an Association Rules can be determined by 3 parameters that are used to identify the strength of the algorithm. Namely –

- Support
- Confidence
- Lift

Let X and Y represent the products in the market and N represents the total number of products.

$$\begin{aligned} \text{Rule: } X \Rightarrow Y & \begin{cases} \text{Support} = \frac{\text{freq}(X, Y)}{N} \\ \text{Confidence} = \frac{\text{freq}(X, Y)}{\text{freq}(X)} \\ \text{Lift} = \frac{\text{Support}}{\text{Supp}(X) \times \text{Supp}(Y)} \end{cases} \end{aligned}$$

Support: It is the probability of an event to occur.

Fraction of transactions that contain an itemset.

For example, the support of item I is defined as the number of transactions containing I divided by the total number of transactions.

$$\text{support}(I) = \frac{\text{Number of transactions containing } I}{\text{Total number of transactions}}$$

Confidence: It is a measure of conditional probability

Measures how often items in Y appear in transactions that contain X

Confidence is the likelihood that item Y is also bought if item X is bought. It's calculated as the number of transactions containing X and Y divided by the number of transactions containing X

$$\text{confidence}(X \rightarrow Y) = \frac{\text{Number of transactions containing } X \text{ and } Y}{\text{Number of transactions containing } X}$$

Lift: It is the probability of all items occurring together divided by the product of antecedent and consequent occurring as if they are independent of each other

These measures show how much more often the antecedent and consequence of the rule occur together than we would expect if they were statistically independent (This measures the strength of the association between X and Y.). A lift value greater than 1 implies that the antecedent and consequent are dependent. (Lift evaluates the strength of the rule over the randomness. A lift greater than 1 indicates a strong rule)

Concept of Itemset –

An 'item' is a collection of one or more items found within a dataset. For example, consider a dataset containing various groceries. An item could be a combination like {Cheese, Tomato}

The 'length' of an item set is the number of items it contains. Thus, {Cheese, Tomato} is a 2-itemset.

- Single item itemsets: {Milk}, {Bread}, {Butter}, {Diapers}, {Sauce}, {Cola}
- Two-item itemsets: {Milk, Bread}, {Bread, Butter}, {Diapers, Sauce}, etc.
- Three-item itemsets: {Milk, Bread, Butter}, {Bread, Diapers, Sauce}, etc

Procedure to implement association rule mining algorithm:

1. **Data Collection:** Use a sample dataset of market transactions. For this experiment, use a grocery store dataset where each row represents a customer's purchase
2. **Data Preprocessing:** Convert the dataset into a transactional format, such as a list of items purchased in each transaction
3. **Implementing association rule mining:** Use the apriori function from the library to find frequent itemsets with minimum support
4. **Generate Association Rules:** Use the association_rules function to generate rules from the frequent itemsets with thresholds for confidence and lift
5. **Evaluate and Analyze results:** Interpret the generated rule and analyze relationships among items

Sample Code:

```
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules
```

Step 1: Data Collection

```
data = {'TID': [1, 2, 3, 4, 5],
        'Milk': [1, 0, 1, 1, 0],
        'Bread': [1, 1, 1, 0, 1],
        'Butter': [0, 1, 0, 1, 1],
```

```
'Cheese': [1, 0, 1, 1, 0]]
df = pd.DataFrame(data).set_index('TID')

print("Dataset:\n", df)

# Step 2: Apply Apriori Algorithm
frequent_itemsets = apriori(df, min_support=0.5, use_colnames=True)
print("\nFrequent Itemsets:\n", frequent_itemsets)

# Step 3: Generate Association Rules
rules = association_rules(frequent_itemsets, metric="confidence",
min_threshold=0.7)
print("\nAssociation Rules:\n", rules[['antecedents', 'consequents',
'support', 'confidence', 'lift']])

# Step 4: Analyzing Results
print("\nStrong Rules (Lift > 1):\n", rules[rules['lift'] > 1])
```

Observations -

- List the frequent itemsets and their respective support values
- Analyze the association rules, noting the confidence and lift values for each rule
- Identify strong rules based on high confidence and lift

Conclusion –

Hence, the model summarizes the key relationships discovered from the dataset and discuss practical applications such as Market Basket Analysis, Recommendation Systems or Inventory Management

References –**a. Textbook –**

- i. Machine Learning with Python – An approach to Applied ML – Abhishek Vijayvargiya, BPB Publications, 1st Edition 2018
- ii. Machine Learning, Tom Mitchell, McGraw Hill Education, 1st Edition 1997

b. Online references –

- i. <https://www.kaggle.com/code/mervetorkan/association-rules-with-python>
- ii. <https://github.com/ymoch/apyori>
- iii. <https://thinkingneuron.com/how-to-do-association-rule-mining-using-apriori-in-python/>

Expected Oral Questions –

1. What is association rule mining?
2. Explain the concepts of “support”, “confidence” and “lift” in association rule mining?
3. What is the difference between frequent itemsets and association rules?
4. Why is support important in association rule mining?
5. What is Apriori algorithm and how does it work?
6. Explain the limitations of the Apriori algorithm
7. What are some common applications of association rule mining in real-world scenarios?
8. What is the purpose of association rule mining?
9. How does association rule mining help in data analysis?
10. What is the difference between frequent itemsets and association rules? How are they related?

FAQ's in Interview –

1. What is the Apriori algorithm?
2. What algorithms can be classified under Association Rule Mining?
3. What are confidence, support and lift?
4. What is Market Basket Analysis?
5. What are the limitations of Association Rule Mining?